

Description of Work (DOW) for:

Support Services: Engineering for IT Environment/Service Desk (SMA3)

1.1 Objectives

CBO requires IT Engineering Support Services to design, implement, and maintain technical controls to reduce the risk of unauthorized initial discovery and lateral movement, malicious credential use and defense evasion, and persistence via machine key and related system-abuse techniques within CBO's environment. The engineering services will also enable secure endpoint, identity, and device lifecycle operations in support of CBO's threat detection and mitigation strategy. This scope does not include routine end-user help desk support, but rather engineering tasks to resolve issues arising from complex or escalated tickets.

The objectives of this effort are:

- Establish and maintain secure, standardized workstation images for macOS and Windows that support both on-site and remote access, including VDI connectivity.
- Implement and maintain endpoint configuration baselines, patching, and version control using approved enterprise tools to reduce configuration drift and security risk.
- Strengthen device registration, enrollment, and asset inventory accuracy to ensure visibility into users, devices, and access across the environment.
- Enhance logging, telemetry, and monitoring capabilities to support detection, investigation, and response to suspicious or unauthorized activity on endpoints and networks.
- Reduce the likelihood and impact of lateral movement and credential misuse through hardened configurations, secure authentication mechanisms, and consistent enforcement of baseline controls.

The Contractor shall provide engineering support by designing and maintaining technical controls, implementing approved configurations, and producing documentation and operational playbooks necessary to support IRM&TS operations, Service Desk engineering escalations, and the incident response function.

1.2 Scope of Work

This document refers to "Engineer" singularly, but it is expected that the Contractor will provide multiple personnel to fill the roles required by this scope of work. The Contractor employees will be responsible for implementing new solutions and leveraging existing tools within the architecture to support logging and auditing of security controls, as well as necessary upgrades and feature enhancements. The team will also assist in administering all information security functions for the Windows and macOS baselines, including updates, upgrades, policy

administration, and validation for special access to CBO's segmented environments (both cloud and non-cloud).

The scope of work includes, but is not limited to, the following activities:

- Assist CBO in building, maintaining, and securing existing standard workstation images that enable access to the VDI environment for remote users on both macOS and Windows platforms.
- Assist CBO in building, maintaining, and securing existing standard workstation images that enable access to the VDI environment for on-site users on both macOS and Windows platforms.
- Maintain operating system and application patching, version control, and lifecycle management for supported applications. Ensure that new and updated applications are delivered through approved mechanisms such as a company application store, Intune, or Group Policy (GPO), as appropriate based on user role and access level.
- Support Microsoft Intune registration and Windows Autopilot for desktops, laptops, and CBO-issued mobile devices, including implementation of passwordless authentication, hardware security keys (e.g., YubiKeys), and other protections for privileged and sensitive accounts.
- Implement and maintain logging, monitoring, and audit capabilities to track device enrollment, user authentication activity, network access, and endpoint behavior to determine which users and devices are connected to the network and what activity is occurring.

1.2.1 Initial Discovery and Strengthening Secure System Posture

The Engineer shall design, implement, and tune engineering controls to detect and limit initial compromise and lateral movement. These activities shall be integrated into workstation imaging, device registration, and user registration processes to ensure secure baseline enforcement and accurate asset inventory management.

The Engineer shall perform the following tasks:

- Design and maintain secure standard workstation images that enable access to the VDI environment for remote users on both macOS and Windows platforms.
- Design and maintain secure standard workstation images that enable access to the VDI environment for on-site users on both macOS and Windows platforms.
- Engineer and maintain operating system and application patching, version control, and lifecycle management for supported applications, ensuring delivery through approved

mechanisms such as a company application store, Microsoft Intune, or Group Policy Objects (GPO) based on user role and access level.

- Engineer and support Microsoft Intune registration and Windows Autopilot for desktops, laptops, and CBO-issued mobile devices, including passwordless authentication and hardware-backed credentials (e.g., YubiKeys, CAC cards, or software-based keys) for privileged and sensitive accounts.
- Engineer and maintain logging, monitoring, and audit capabilities to track device enrollment, user authentication, network access, and endpoint activity to determine which users and devices are connected to the network and what activity is occurring.

1.2.2 Endpoint Configuration, Baseline, and Device Lifecycle Management

The Engineer shall design, implement, and maintain standardized endpoint configurations to ensure the security, compliance, and integrity of user devices and associated assets throughout their lifecycle. This work shall be performed using approved endpoint management and deployment tools, including Ivanti, KACE, Microsoft Intune, and Windows Autopilot, as applicable.

Responsibilities include, but are not limited to:

- Engineering and maintaining secure baseline configurations for macOS and Windows endpoints to support both on-site and remote access use cases.
- Using Ivanti and KACE to manage operating system and application patching, version control, deployment workflows, and remediation of configuration drift.
- Leveraging Microsoft Intune to enforce device compliance, configuration profiles, security policies, and conditional access requirements based on user role and device posture.
- Supporting Windows Autopilot for automated device provisioning, registration, and lifecycle management to ensure devices are securely configured at first use.
- Integrating device registration and user association processes to ensure accurate asset inventory, ownership tracking, and lifecycle visibility.
- Ensuring that endpoint configurations support secure authentication methods, including passwordless authentication and hardware-backed credentials, where approved.
- Monitoring endpoint configuration and compliance status to identify deviations from approved baselines and implementing corrective actions as necessary.
- Documenting baseline standards, deployment procedures, and remediation workflows to support operational continuity and audit requirements.

1.2.3 Imaging, Patching, and Automation (Engineering)

The Engineer shall own engineering design, validation, and lifecycle maintenance of imaging and patching processes to ensure consistent, secure workstation builds and timely vulnerability remediation. The Engineer shall assess and harden machine identity artifacts, engineer detection for persistence mechanisms, and develop remediation and recovery runbooks.

Responsibilities include:

- Design, build, and maintain standardized, division-specific workstation images for macOS and Windows that incorporate approved baseline security controls and required VDI/remote access clients.
- Maintain imaging toolchains and automation scripts (Ivanti, KACE, JAMF, or equivalent) used for image creation, testing, and deployment; validate image integrity prior to production release.
- Engineer and operate patch management processes using Ivanti (or equivalent) for OS and third-party application patching; coordinate Intune/GPO-based patch orchestration for Windows endpoints.
- Implement automation for image deployment, patch orchestration, post-patch validation, and rollback procedures to minimize manual intervention and reduce MTTR.
- Maintain version control and a formal image-release process (build → test → signoff → release); publish image versions and changes to the COR and affected stakeholder groups.
- Develop and execute validation tests after imaging or major patch cycles to confirm endpoint functionality (connectivity to VDI, authentication, application compatibility, and security agents).
- Document imaging and patch runbooks, recovery procedures, and rollback steps; retain change records in the official project repository.

1.2.4 Device Enrollment, Provisioning, and Asset Management

The Engineer shall ensure secure, consistent device enrollment and accurate asset tracking across the estate. The Contractor shall produce engineering documentation, operational playbooks, and conduct knowledge transfer sessions.

Responsibilities include:

- Implement and maintain enrollment workflows for Intune, Autopilot (Windows), and Apple Business Manager / JAMF for macOS and iOS devices.

- Ensure device provisioning enforces baseline profiles and security posture prior to granting network access (conditional access integration with Entra/AD).
- Integrate enrollment and provisioning with asset-inventory systems so that each device is associated with owner, role, location, and lifecycle state.
- Support lifecycle operations: provisioning, reassignment, decommission, and secure wipe; ensure proper escrow of recovery keys (e.g., FileVault) and custody records.
- Maintain an onboarding/offboarding checklist and automate steps where feasible to reduce human error and accelerate device readiness.

1.2.5 Monitoring, Logging, and Audit Engineering

The Engineer shall ensure endpoints produce the telemetry required for detection, troubleshooting, and forensics and that telemetry flows to central systems in a reliable and auditable manner.

Responsibilities include:

- Ensure endpoint agents (EDR, AV, logging agents) are installed and reporting, and that logs (Windows Event, macOS Unified Logs, application logs) are forwarded to the SIEM/EDR platform according to documented schemas.
- Configure and maintain log-forwarding, parsing, and normalization rules so that security-relevant events are usable for detection and incident response.
- Implement monitoring of enrollment, imaging, patch status, and compliance posture; publish alert thresholds for operational failures (e.g., image deployment failures, patch rollback events, agent offline).
- Provide engineering support for forensic collections when required (artifact collection playbooks, preservation steps, and non-production test validations).
- Maintain audit trails of configuration changes, image releases, and remediation actions for compliance and post-incident review.

1.2.6 Assessment → Recommendation → Implementation Workflow

All work will be assigned by the COR following an assessment-to-remediation workflow. All change activities shall be documented and retained; emergency changes shall follow expedited approval processes agreed with the COR. To ensure findings produce fixes, the Engineer shall follow a formal workflow:

1. Assess: Perform technical assessment; produce a Findings Report that captures risk, impacted assets, and priority.

2. Recommend: Submit a written Remediation Plan with implementation steps, validation criteria, rollback procedures, resource needs, and proposed change window.
3. Approve: Obtain COR and CO approvals as required by CBO change control and procurement policies.
4. Implement: Execute the approved remediation during the agreed change window with full documentation of changes.
5. Validate: Perform post-change validation and publish a Validation Report.

1.2.7 Knowledge Transfer, Documentation, and Runbooks

The Engineer shall make solutions operationally sustainable by the Service Desk and IRM teams.

Responsibilities:

- Produce user-facing runbooks for imaging/recovery, patch-validation steps, enrollment troubleshooting, and common remediation tasks that the Service Desk can safely execute.
- Conduct periodic training and working sessions with Service Desk leads, IAM, and SOC staff to review new baselines, automation, and outage recovery steps.
- Maintain a living knowledge-base with versioned documentation and clear escalation paths for image/patch/enrollment failures.

1.2.8 Exclusions and Boundaries

- The Engineer shall not provide routine Tier 1 help desk ticket handling or general end-user desktop support unrelated to imaging, baseline remediation, enrollment, patching, or telemetry engineering.
- The Engineer shall not be the final authority for security policy decisions; technical implementations will be coordinated with and approved by the COR, Cybersecurity leadership, and the CO as required.

1.2.8.1 Engineering Tooling and Software Procurement Support

The Contractor may identify the need for software, licenses, tools, or subscriptions required to perform the engineering tasks defined in this Description of Work (e.g., imaging tools, automation modules, logging agents, or endpoint management extensions).

- The Contractor shall not independently procure items or spend government funds without prior written CO approval.
- The Contractor may submit written recommendations and justifications to the COR identifying required tools, licensing models, estimated costs, and implementation impact.

- All acquisitions shall be reviewed and approved by the CO in accordance with CBO procurement policies prior to purchase.
- Upon approval, the Contractor may assist with implementation, configuration, and validation of approved tools.

1.3 Deliverables and Acceptance

The Contractor shall provide the following deliverables in support of the engineering tasks defined in Sections 1.2.1 through 1.2.8. Deliverables will be reviewed and accepted by the Contracting Officer's Representative (COR).

Deliverable	Associated Section(s)	Description / Acceptance Criteria
Secure Workstation Image Baselines (macOS and Windows)	1.2.1, 1.2.3	Images validated for on-site and remote/VDI access; security agents installed; authentication and logging verified
Image Version Control and Change Documentation	1.2.1, 1.2.3	Versioned image history, change summary, rollback steps documented
Patch and Application Deployment Summary	1.2.1, 1.2.2, 1.2.3	OS/app patch status, deployment method (Ivanti/KACE/Intune/GPO), validation results
Device Enrollment and Authentication Configuration Summary	1.2.1, 1.2.2, 1.2.4	Intune/Autopilot/JAMF enrollment flows, authentication methods, enforcement status
Endpoint Telemetry and Logging Coverage Summary	1.2.1, 1.2.5	Log sources, destinations, health status, audit coverage
Endpoint Baseline Configuration Standards	1.2.2	Documented macOS/Windows baselines suitable for audit
Imaging, Patching, and Automation Runbooks	1.2.3	Build, patch, validation, rollback, automation scripts
Device Lifecycle Management Workflow Documentation	1.2.2, 1.2.4	Provisioning, reassignment, decommissioning workflows
Asset Inventory Reconciliation Summary	1.2.2, 1.2.4	Device-to-user accuracy, discrepancies resolved
Onboarding/Offboarding Checklist	1.2.4	
Telemetry Health Report	1.2.5	Monthly or as directed
Incident Forensic Playbook (appendix)	1.2.5	
Findings Report, Remediation Plan, Validation Report, Change Log	1.2.6	Assessment → remediation lifecycle evidence

Deliverable	Associated Section(s)	Description / Acceptance Criteria
Knowledge Transfer Materials Runbooks set (image, patch, enrollment) and Attendance Logs	1.2.7	Training artifacts and participation, Quarterly KT schedule

1.3 Key Personnel & Requirements

The following roles are anticipated to be required for this work and are considered Key Personnel, in accordance with SOW paragraph 2.9.

- Engineer
- Cloud Service Desk Rep
- Apple/macOS SME
- Microsoft SME

The Contractor shall provide personnel with demonstrated engineering expertise appropriate to the tools and responsibilities in this Description of Work and who meet the requirements of this Section 1.3.

1.3.1 Core Engineering Experience (Required for All Roles)

- Eight (8) years of experience in Information Technology, Endpoint Engineering, or Cybersecurity.
- Six (6) years of experience performing **engineering (not help desk)** functions in enterprise environments.
- Experience working under formal change control, audit, and security governance processes.

1.3.2 Endpoint Imaging and Automation Engineering

Applies to Sections 1.2.1 and 1.2.3.

- Experience building and maintaining **Windows and macOS workstation images**.
- Experience with image automation, validation, rollback, and version control.
- Experience integrating images with VDI, EDR, authentication, and logging agents.

1.3.3 Patch and Configuration Management Engineering (Ivanti / KACE)

Applies to Sections 1.2.1, 1.2.2, and 1.2.3.

- Hands-on experience using Ivanti and/or KACE for OS and application patching.
- Experience managing configuration drift, remediation workflows, and reporting.

- Experience validating patches post-deployment and supporting rollback.

1.3.4 Device Enrollment and Identity Engineering (Intune / Autopilot / JAMF)

Applies to Sections 1.2.1, 1.2.2, and 1.2.4.

- Experience with Microsoft Intune and Windows Autopilot for provisioning and compliance enforcement.
- Experience using JAMF Pro for macOS endpoint management (macOS only; not iOS unless directed).
- Experience implementing passwordless authentication and hardware-backed credentials (YubiKey, CAC, software keys).

1.3.5 Logging, Monitoring, and Telemetry Engineering

Applies to Sections 1.2.5 and 1.2.6.

- Experience configuring endpoint logging (Windows Event Logs, macOS Unified Logs).
- Experience forwarding and validating logs into SIEM/EDR platforms such as MS Sentinel (not limited).
- Experience supporting forensic collection and audit readiness.

Preferred Education:

Bachelor's degree in Information Technology, Cybersecurity, or a related field (or equivalent experience).

Education Substitution:

Formal education requirements may be waived based on relevant professional experience, as determined by the COR.

1.3.2 Required Credentials

The candidate(s) provided by the Contractor must have the following current credentials:

- Top Secret (TS) security clearance

1.4 Schedule of Work

CBO anticipates that the Contractor employees will generally work up to 40 hours per week at the applicable hourly rates specified in the price schedule, excluding federal holidays or days the agency is closed. Some tasks may be limited in duration and performed on an as-needed basis for initial setup, configuration, and validation, and may not require continued reach-back support.